

BTEC Assignment Brief

Qualification	Pearson BTEC Level 5 Higher Nationals in Digital Technologies
Unit number and title	Unit 36: Application Development
Learning aim(s)	By the end of this unit students will be able to: LO1. Produce a Software Design Document by analysing a business-related problem and deduce an appropriate solution including a set of initial requirements. LO2. Use design and development methodologies with tools and techniques associated with the creation of a business application. LO3. Work individually and as part of a team to plan and produce a functional business application with support documentation. LO4. Evaluate the performance of a business application against its Software Design Document and initial requirements.
Assignment title	Development, Testing, Deployment and Evaluation of a Business Application
Assessor	Javokhir Fayzullayev
Issue date	
Hand in deadline	

Context	Students will continue from their approved Software Design Document and progress to full development and implementation. Working individually or as part of a small development team, students are required to build a functional business application that meets the identified system and user requirements. Students must apply appropriate development methodologies, tools, testing strategies and documentation practices in order to successfully deliver and evaluate the application.
Task	Students must: • Plan, prepare and develop a functional business application • Implement the application using appropriate tools, programming techniques and development methodology • Apply appropriate algorithms, logic and data structures • Produce technical and user documentation • Create testing documentation (test plan, test cases, results) • Conduct comprehensive testing and fix identified defects • Deploy the application (or provide clear deployment instructions) • Evaluate application performance against: • original requirements • software design document • Reflect on development activities, decisions and outcomes
Evidence Required	Students must submit: • Fully functional business application • Source code / repository link / installation package • User Guide • Technical / Developer Documentation • Testing documentation (plan + results) • Evaluation report (performance + reflection)

Learning Outcomes and Assessment Criteria

Pass		Merit	Distinction
LO1 Produce a Software Design Document by analysing a business-related problem and deduce an appropriate solution including a set of initial requirements.			LO1 & LO2 D1 Justify your solution to a business-related problem and your preferred software development methodology, by comparing between the various software development tools and techniques researched.
P1 Explore a business-related problem and produce a well-defined Problem Definition Statement supported by a set of user and system requirements.	P2 Determine any areas of risk related to the successful completion of your application.	M1 Analyse a business-related problem using appropriate methods and produce a well-structured Software Design Document that defines a proposed solution and includes relevant details on requirements, system analysis, system design, coding, testing and implementation.	
LO2 Use design and development methodologies with tools and techniques associated with the creation of a business application			
P3 Research the use of software development tools and techniques and identify any that have been selected for the development of this application		M2 Compare the differences between the various software development tools and techniques researched and justify your preferred selection as well as your preferred software development methodology	
LO3 Work individually and as part of a team to plan and produce a functional business application with support documentation			D2 Evaluate any new insights, ideas or potential improvements to your system and justify the reasons why you have chosen to include (or not to include) them as part of this business application.
P4 Create a formal presentation that effectively reviews your business application, problem definition statement, proposed solution and development strategy. Use this presentation as part of a peer-review and document any feedback given.	P5 Develop a functional business application with support documentation based on a specified business problem.	M3 Interpret your peerreview feedback and identify opportunities not previously considered. M4 Develop a functional business application based on a specific Software Design Document with supportive evidence of using the preferred tools, techniques and methodologies.	
LO4 Evaluate the performance of a business application against its Software Design Document and initial requirements			

P6 Review the performance of your business application against the Problem Definition Statement and initial requirements.	M5 Analyse the factors that influence the performance of a business application and use them to undertake a critical review of the design, development and testing stages of your application. Conclude your review by reflectively discussing your previously identified risks.	D3 Critically evaluate the strengths and weaknesses of your business application and fully justify opportunities for improvement and further development.
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